

Requirements Document

Functional Requirements

1. Drones must not enter no-fly zones and restricted zones.
2. Drones must not cross the corners of no-fly zones and restricted zones
3. Drones must be dispatched from central static drone points and return to the same drone point.
4. Drones must be fully charged when arriving at a drone service point.
5. Drones must only accept deliveries that do not exceed the drone's maximum moves and maximum cost.
6. Drones can only move in one of the 16 compass directions, each 22.5° apart (N, NE, E, SE, S, SW, W, NW, NNE, ENE, ESE, SSE, SSW, WSW, WNW, NNW).
7. Each drone move must be exactly 0.00015° in length.
8. Delivery locations must be specified using longitude and latitude coordinates.
9. Drones must hover once upon reaching a delivery location.
10. System must check for drone availability before dispatching drones.
11. Drones can only accept orders if they meet the order's heating and cooling requirements.
12. System must fetch drone and service point data from the external ILP Rest Service for each dispatch order.

Measurable Quality Attributes

1. Total time taken to calculate drone path must not exceed 200 milliseconds.
2. Total flight cost = (costInitial + costFinal) + (moves $\times$ costPerMove) must be minimized.
3. Total moves drones use must be minimized.
4. Drone positions must be accurate to within $\pm 10^{-12}^\circ$ of the calculated point.
5. System must support NFZ databases of up to 1000 NFZ.

Qualitative Requirements

1. The mobile user-interface of the app should be user-friendly.
2. Code should be readable and understandable for future developers.
3. Error messages should be clear and helpful.
4. Code architecture should follow clear separation of concerns (service, controllers, DTOs remain separate).